A NINE-ELEMENT COUNTEREXAMPLE TO SINCLAIR'S FRONT-LOADING CONJECTURE

Abstract. Sinclair's Conjecture C asserts that the defect of every simple matroid is front-loaded. We give a simple rank-five matroid M on nine elements, representable over $\mathbb{Q}$, whose apolar Hilbert series and radial-derivative Hilbert series, both recalled in Section 1, are

$$H^0_M(q) = 1 + 9q + 27q^2 + 27q^3 + 9q^4 + q^5$$

and

$$H_{\beta,M}(q) = 1 + 9q + 31q^2 + 32q^3 + 9q^4 + q^5.$$

Here $h^\beta_k(M)$ and $h^0_k(M)$ denote the coefficients of q^k in $H_{\beta,M}$ and in H^0_M, so the defect $\delta_k(M) = h^\beta_k(M) - h^0_k(M)$ has $\delta_2(M) = 4 < 5 = \delta_3(M)$, which refutes the conjecture. The calculation uses exact rational arithmetic and is reproduced by the short standard-library program printed in the appendix.

1. The counterexample

We use Sinclair's notation [1]. Let M be a simple matroid of rank r on a ground set E, and let g_E be the top class of its graded Möbius algebra, rescaled as in the proof of Theorem 1: there $g_X = \beta_X e_X$ for a flat X, where β_X is the number of ordered bases of X and e_X is the corresponding basis vector of the graded Möbius algebra. Write

$$H_{\beta,M}(q) = \sum_{k=0}^{r} h^\beta_k(M) q^k$$

for the Hilbert series of the cyclic module generated by g_E under the radial derivative D_β and the coordinate derivatives L^β_a ($a \in E$), graded by the number of operator applications: in the notation of the chain (2) below, $h^\beta_k(M) = \dim_{\mathbb{Q}} \mathcal{C}^\beta_k$. Omitting D_β gives the classical apolar Hilbert function $h^0_k(M) = \dim_{\mathbb{Q}} \mathcal{C}^0_k$, and we write

$$H^0_M(q) = \sum_{k=0}^{r} h^0_k(M) q^k.$$

Sinclair's Conjecture C states that, with

$$\delta_k(M) = h^\beta_k(M) - h^0_k(M),$$

one has

$$\delta_k(M) \geq \delta_{r-k}(M), \qquad k \leq r/2.$$

2020 *Mathematics Subject Classification.* 05B35.

Key words and phrases. matroid, graded Möbius algebra, differential Hilbert function, counterexample.

Sinclair records the equivalent form $h^\beta_k(M) \geq h^\beta_{r-k}(M)$ for $k \leq r/2$ [1]; the two forms agree for the matroid of Theorem 1, whose h^0_k is symmetric under $k \mapsto r-k$. The introduction of [1] describes the same inequality as top-heaviness of H_β in differential degree.

Theorem 1. *Let M be the column matroid over $\mathbb{Q}$ of*

$$A = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 & 1 & 2 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 1 & 3 & 1 \end{pmatrix}.$$

Then M is a simple rank-five matroid on nine elements and

$$H^0_M(q) = 1 + 9q + 27q^2 + 27q^3 + 9q^4 + q^5,$$
$$H_{\beta,M}(q) = 1 + 9q + 31q^2 + 32q^3 + 9q^4 + q^5.$$

Consequently, M is a counterexample to Conjecture C of Sinclair [1].

Proof. The first five columns of A form the identity matrix, so $\mathrm{rk}(M) = 5$. Every column is nonzero and no two columns are proportional; hence M is simple.

For a flat X, let β_X be the number of ordered bases of X, let e_X denote the corresponding basis vector of the graded Möbius algebra, and put $g_X = \beta_X e_X$. By Sinclair's operator formula [1, Proposition 2.4],

$$L^\beta_a g_Y = \sum_{\substack{X \lessdot Y \\ X \vee a = Y}} g_X, \qquad D_\beta g_Y = \mathrm{rk}(Y) \sum_{X \lessdot Y} \frac{|Y \setminus X|}{9 - |X|}\, g_X. \tag{1}$$

Indeed, the atoms are the elements of E, and for a cover $X \lessdot Y$ one has $X \vee a = Y$ exactly when $a \in Y \setminus X$. Thus the quantities in Sinclair's formula satisfy

$$s(X) = 9 - |X|, \qquad m(X, Y) = |Y \setminus X|.$$

Both are evaluated at the lower flat of the cover: the sets $Y \setminus X$, as Y runs over the covers of X, partition the atoms not lying below X, so that $\sum_{Y \gtrdot X} m(X, Y) = 9 - |X| = s(X)$.

Starting from $\mathcal{C}^\beta_0 = \mathbb{Q} g_E$, define

$$\mathcal{C}^\beta_{k+1} = D_\beta \mathcal{C}^\beta_k + \sum_{a \in E} L^\beta_a \mathcal{C}^\beta_k, \tag{2}$$

and define $\mathcal{C}^0_k$ by omitting D_β; as in Section 1, $h^0_k(M) = \dim_{\mathbb{Q}} \mathcal{C}^0_k$ and $h^\beta_k(M) = \dim_{\mathbb{Q}} \mathcal{C}^\beta_k$, which are the numbers tabulated below. All matrices in (1) have rational entries, so their ranks over $\mathbb{Q}$ agree with their ranks over $\mathbb{R}$.

For each $S \subseteq E$, exact elimination computes $\mathrm{rk}(A_S)$ and hence

$$\mathrm{cl}(S) = \{a \in E : \mathrm{rk}(A_{S \cup \{a\}}) = \mathrm{rk}(A_S)\}.$$

Enumerating the 2^9 subsets gives the Whitney numbers

$$(W_0, W_1, W_2, W_3, W_4, W_5) = (1, 9, 32, 49, 27, 1).$$

Constructing the flat-cover matrices in (1) and applying exact Gauss–Jordan elimination degree by degree gives

k	0	1	2	3	4	5
$W_{5-k}(M)$	1	27	49	32	9	1
$h^0_k(M)$	1	9	27	27	9	1
$h^\beta_k(M)$	1	9	31	32	9	1
$\delta_k(M)$	0	0	4	5	0	0

For $r = 5$, the index $k = 2$ lies in the conjectured range, but

$$h^\beta_2(M) = 31 < 32 = h^\beta_3(M), \qquad \delta_2(M) = 4 < 5 = \delta_3(M).$$

This refutes Conjecture C. □

Remark 2. The sequence $(1, 9, 31, 32, 9, 1)$ is strictly log-concave, so the example does not contradict Sinclair's separate log-concavity conjecture. No claim of minimality is made.

Status. Conjecture C fails for M in both of the forms printed in [1], and only in the range $k \leq r/2$ that the conjecture addresses; the cases proved there — the endpoints $\delta_0 = \delta_r = 0$ and the uniform matroids, where $H_\beta = H^0$ — are untouched, and M is neither. As we read [1], Conjecture C is verified there by exact rational elimination for all 950 pairwise nonisomorphic simple matroids on eight elements [1, Prop. 5.1], and a modular check of a stratified sample of 400 simple nine-element matroids is reported [1, Sec. 5] and called there evidence rather than an exact certificate; that sample is specified by a seed, and we make no claim about whether M lies in it. We have not reverified either of those computations, and Theorem 1 does not depend on them: they bear only on where a counterexample could already have been found. On the same reading, no exhaustive nine-element census is asserted in [1], and we are not aware of any other work that resolves Conjecture C or exhibits a counterexample to it; the invariants H_β, D_β and δ_k are introduced in [1]. Every statement about [1] in this paragraph is a reading of the version cited below, not an output of the computation reported next.

Exact verification

The computation is a direct enumeration in exact rational arithmetic. All 2^9 subsets of E are ranked by elimination over $\mathbb{Q}$, which yields the closure operator, the 119 flats, the Whitney numbers and the covering relations; the matrices of (1) are then assembled on the basis $\{g_X\}$ and the chain (2) is reduced one degree at a time. The program in Appendix A carries this out with the Python standard library alone, every arithmetic operation performed on `Fraction` objects, and prints the rank and simplicity of M, its number of bases (67), its number of flats (119), the Whitney numbers, the join step used in the proof above, both Hilbert functions and δ. No floating-point computation and no external matroid catalogue is used.

The same assertions were recomputed independently on a separate machine, by a longer program written for this check from the statements above rather than from the listing below, again using the Python standard library alone and exact arithmetic throughout. It reports 72 checks and all 72 pass; it agrees with every number printed here, namely rank five, simplicity, 67 bases, 119 flats, $(1, 9, 32, 49, 27, 1)$, $(1, 9, 27, 27, 9, 1)$, $(1, 9, 31, 32, 9, 1)$ and $(0, 0, 4, 5, 0, 0)$, and it brackets each dimension from both sides, by a non-vanishing minor of that size and by explicit annihilating functionals. As a control it was run on the uniform matroid $U_{5,9}$, which is determined up to isomorphism by its rank and its number of elements, and there it reports that Conjecture C holds; so the test is not one that fires on every input. The first program is printed in full in Appendix A, and the longer program together with the transcript of its run accompanies this note as supporting material.

Two programs agreeing is not a proof, and two things lie outside the reach of both, each being a matter of reading [1] rather than of computation: the transcription of the operator formula (1), and the convention that δ_k is indexed by the number of operator applications, as in (2) and in the table above. Both are printed in full above rather than only cited — (1) together with the identifications $s(X) = 9 - |X|$ and $m(X, Y) = |Y \setminus X|$ that specialize it to M, and the grading in (2) — so that they can be compared with the source directly. The Hilbert functions of Theorem 1 are exact consequences of the definitions of Section 1 and do not depend on that comparison; the identification of the inequality they violate with Conjecture C does. The grading is the load-bearing half: indexing instead by the degree of the graded piece, $h_j = \dim_{\mathbb{Q}} \mathcal{C}_{r-j}$, replaces δ by $(0, 0, 5, 4, 0, 0)$, which satisfies the inequality. There is no census or minimality claim here, and hence no exhaustive search to reproduce.

Appendix A. The verification program

The following program is self-contained and recomputes every number in the table of Theorem 1 from the matrix A alone. Comments give the expected output, one `print` statement per line.

```
from fractions import Fraction as F
from itertools import combinations
from math import factorial

A = [[1, 0, 0, 0, 0, 1, 0, 0, 0],
     [0, 1, 0, 0, 0, 0, 1, 1, 0],
     [0, 0, 1, 0, 0, 1, 1, 2, 0],
     [0, 0, 0, 1, 0, 0, 0, 0, 1],
     [0, 0, 0, 0, 1, 0, 1, 3, 1]]
```

```
n, r = 9, 5
E = frozenset(range(n))

def rref(rows):                    # exact Gauss-Jordan; returns
    M = [list(v) for v in rows]   # the nonzero reduced rows
    p = 0
    for c in range(len(M[0])):
        piv = next((i for i in range(p, len(M))
                    if M[i][c] != 0), None)
        if piv is None:
            continue
        M[p], M[piv] = M[piv], M[p]
        M[p] = [x / M[p][c] for x in M[p]]
        for i in range(len(M)):
            if i != p and M[i][c] != 0:
                f = M[i][c]
                M[i] = [x - f * y for x, y in zip(M[i], M[p])]
        p += 1
    return M[:p]

def rank_cols(S):
    S = sorted(S)
    return len(rref([[F(A[i][c]) for c in S]
                     for i in range(r)])) if S else 0

rk = {frozenset(S): rank_cols(S)
      for k in range(n + 1) for S in combinations(range(n), k)}
print(rk[E],
      [a for a in range(n) if rk[frozenset([a])] == 0],
      [p for p in combinations(range(n), 2)
       if rk[frozenset(p)] == 1])
#      5 [] []                       rank five and simple
print(sum(1 for S in combinations(range(n), r)
          if rk[frozenset(S)] == r))
#      67                            bases

def cl(S):
    S = frozenset(S)
    return frozenset(a for a in range(n) if rk[S | {a}] == rk[S])

flats = sorted({cl(S) for S in rk},
               key=lambda X: (rk[X], sorted(X)))
lev = {k: [X for X in flats if rk[X] == k] for k in range(r + 1)}
pos = {X: i for k in lev for i, X in enumerate(lev[k])}
print(len(flats), [len(lev[k]) for k in range(r + 1)])
#      119 [1, 9, 32, 49, 27, 1]    flats, then the Whitney numbers

und = {Y: [X for X in lev[rk[Y] - 1] if X < Y]   # covers X < Y
```

```
       for k in range(1, r + 1) for Y in lev[k]}
print(all((cl(X | {a}) == Y) == (a in Y - X)
          for Y in und for X in und[Y] for a in range(n)))
#      True                          the join step used in the proof

def beta(X):                    # ordered bases of the flat X
    d = rk[X]                   # beta(E) = 8040 = 67 * 5!
    return factorial(d) * sum(1 for S in combinations(sorted(X), d)
                              if rk[frozenset(S)] == d)

def L(v, k, a):                 # v is indexed by lev[k]
    out = [F(0)] * len(lev[k - 1])
    for Y, c in zip(lev[k], v):
        for X in (und[Y] if c else []):
            if cl(X | {a}) == Y:
                out[pos[X]] += c
    return out

def D(v, k):
    out = [F(0)] * len(lev[k - 1])
    for Y, c in zip(lev[k], v):
        for X in (und[Y] if c else []):
            out[pos[X]] += c * rk[Y] * F(len(Y - X), n - len(X))
    return out

def hilb(use_D):
    v, k = [[F(beta(E))]], r    # C_0 = Q g_E
    h = [len(rref(v))]
    while k:
        nxt = ([D(u, k) for u in v] if use_D else [])
        nxt += [L(u, k, a) for u in v for a in range(n)]
        v = rref(nxt)
        h.append(len(v))
        k -= 1
    return h

h0, hb = hilb(False), hilb(True)
print(h0, hb, [b - a for a, b in zip(h0, hb)])
#      [1, 9, 27, 27, 9, 1] [1, 9, 31, 32, 9, 1] [0, 0, 4, 5, 0, 0]
print(hb[2] < hb[3], hb[2] - h0[2] < hb[3] - h0[3], 2 <= r / 2)
#      True True True                Conjecture C fails at k = 2
```

References

[1] T. Sinclair, *The radial derivative on the graded Möbius algebra*, arXiv:2608.18519v1 [math.CO], 2026.